FLOOD INTELLIGENCE SYSTEM

From unstructured flood reporting to evidence-backed decision support

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4,742
FLOOD-RELATED RECORDS

2-STAGE
LLM WORKFLOW

EVENT → INCIDENT → IMPACT
STRUCTURED INTELLIGENCE

01 HOW THE SYSTEM WORKS

SOURCE NEWS
Historical flood articles from multiple sources

→

EVENT VERIFICATION
LLM checks article relevance to flood events

→

AI EXTRACTION
Incidents, impacts, dates, locations & evidence IDs

→

POSTGRESQL
Structured, queryable evidence-linked records

→

INSIGHTS
Interactive dashboard + event-level RAG

02 STRUCTURED IMPACT KNOWLEDGE

One event. Many incidents. Evidence-backed consequences.

EVENT	Cyclone / flood episode
INCIDENT	River flooding · slips · road flooding
IMPACT	Evacuations · transport · property
EVIDENCE	Source article links + locations

03 POLICY-READY DASHBOARD & EVIDENCE REVIEW



Geospatial impact hotspots



Affected regions · impact categories · trends

FUTURE DIRECTION Connect historical impacts with flood forecasts for impact-based warnings: what could be affected, not only where flooding may occur.